Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Instructions to authors for the preparation of
ISARC manuscripts

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This document sets out the requirements for preparing manuscripts for the ISARC. It is essential that all manuscripts conform to these instructions. Note that this document contains both general instructions and structures specific to the preparation of manuscripts using LaTeX. For instructions on how to prepare manuscripts using MS Word, please use the MS Word template.

Instructions; Formatting; Authors; ISARC

1 Introduction

This LaTeX template and the instructions it contains will enable you to prepare your manuscript in an electronic format (pdf), ready for submission and peer review. It is therefore essential that these instructions be carefully followed.

Your manuscript must be submitted to our online peer-review management system.

The recommended length for Scientific Papers is 6 to 8 pages (4 pages for Short Papers), including text, figures, tables, and references. For more information about the paper types, refer to the "Paper and Presentation Types" on the IAARC website. Electronic files should not exceed 5 megabytes.

Authors are responsible for ensuring the accuracy of all information contained in their manuscripts (e.g., proper names of organizations, data and findings, references, etc.). Manuscripts must be prepared in proper English. SI units or acceptable metric equivalents must be used throughout.

2 Preparation of the manuscript

The manuscript shall be prepared using either the MS Word (.docx) template or LaTeX using the Overleaf template available on the IAARC website. You can alternatively download the source code from the Overleaf template and use it in another editor of your choice.

The rest of the manuscript details the formatting requirements for your submission.

Note that this document is a template, and thus already implements all the specified formatting. Therefore, if you use it to write your manuscript, and utilize the same commands, then your manuscript will have the correct format.

2.1 Page Size and Layout

Your manuscript must be prepared for A4-size (210 x 297 mm) paper. Use the margin settings specified in Table 1 and do not number the pages of the paper. The LaTeX template already implements those margins.

2.1.1 Headers and footers

Do not add anything in the footers or headers, even page numbers. These will be added by the IAARC Technical Committee when compiling the Proceedings.

Tabel 1: Manuscript margins

Margin	A4 (210 x 297 mm)
Top	3.5 cm
Bottom	3.5 cm
Left	2.5 cm
Right	2.0 cm

2.2 Type Style and Spacing

All text must use the Times New Roman font. This LaTeX template already implements this.

2.2.1 Title Page

The title page of your manuscript is illustrated by the first page of these instructions. The title page must not exceed one (1) page in length. It contains:

[noitemsep]The title of the paper in bold 18 points Times New Roman, centered with first letters in capitals. The authors' names, separated from the title by a one blank line, are centered and in size 11, bold Times New Roman. The authors' affiliations and addresses are put immediately below the names, centered and single-spaced, in size 10. Email addresses are inserted below the affiliations, also in size 10.

This template already implements all these requirements. Simply use the commands `\title{}`, `\author{}`, `\affiliation{}` and `\email{}` as illustrated here.

2.2.2 Body of paper

The body of the paper follows the front matter. It contains two columns (with 0.5cm separation) and begins with:

[noitemsep]The major heading "Abstract –", in bold. The body of the abstract, not to exceed 250 words in length, in bold Times New Roman, fully justified, the first line is indented. The major heading "Keywords –" in bold, separated from the last line of the abstract by one blank line. The list of keywords, not to exceed ten (10) words, in bold Times New Roman, left justified, indented, and separated by commas. Please add those keywords that you would use if you were searching for your paper.

The main text of the paper begins after. Separate sections of the main text in accordance with the Headings guidelines below

This template already implements all these requirements. Simply use the environments `abstract` and `keywords` as illustrated in this template.

2.2.3 Headings

All headings must be in black and in bold face. The manuscript will typically have maximum three levels of headings maximum: Level 1, Level 2, and Level 3. Level-1 headings (e.g. Introduction, Background, Discussion, Conclusions, Acknowledgments, References) have font size 12; Level-2 headings have font size 11; and Level-3 headings have font size 10.

Words in the headings should be all lower case, except the first letter of the first word which must start with a capital letter. The headings are positioned at the left margin. They are numbered in the form "1." for Level-1 headings, "1.1." for Level-2 headings, and "1.1.1" for Level-3 headings.

The commands `\section{}`, `\subsection{}` and `\subsubsection{}` employed in this template implement the requirements above for headings Level 1, 2 and 3. Note that you may also use the command `\paragraph{}`, if this suits better your needs.

2.2.4 Text

Text paragraphs are single-spaced and fully justified, with the first line indented 0.5 cm. Do not use blank lines between paragraphs unless you feel it important to really highlight a change of topic (in such a case, you should also consider using sub-headings).

Note that, to start a new paragraph in LaTeX, simply leave a blankline in the `.tex` file.

2.2.5 Lists

[noitemsep]Numbered lists should be presented using the environment `enumerate` like in this example. This environment applies the numbering and defines the format and spacing automatically. Add the option `[noitemsep]` to the environment to have no additional space between numbered list items.

[noitemsep]Bulleted lists should be presented using the environment `itemize` like in this example. This environment applies the numbering and defines the format and spacing automatically. Add the option `[noitemsep]` to the environment to have no additional space between bulleted list items.

Note that the two list above were generated using the environments and options described within them.

2.2.6 Footnotes

Do not use footnotes. Incorporate all required information in the body of the paper.

2.2.7 Equations and symbols

Simple mathematical expressions and sub- and super-scripted characters, such as SO_4^2, are inserted in the text. Do not embed equations as an image. LaTeX is valued particularly for its performance in handling equations and symbols.

Equations are placed on separate lines, centered and numbered consecutively in parentheses at the right-hand margin. A blank line precedes and follows each equation.

To achieve this, simply use the environment *'equation'* as in the examples below.

$$ZnS + 3/2O_2 \rightarrow ZnO + SO_2 \tag{1}$$

$$ZnS + 3/2O_2 = ZnO + SO_2 \tag{2}$$

$$E = 1.23 - 0.06pH \tag{3}$$

The nomenclature and units for symbols must be defined in the text. Alternatively, where the number of symbols is large, a special Section **Nomenclature** should be used at the end of the manuscript.

2.3 Figures

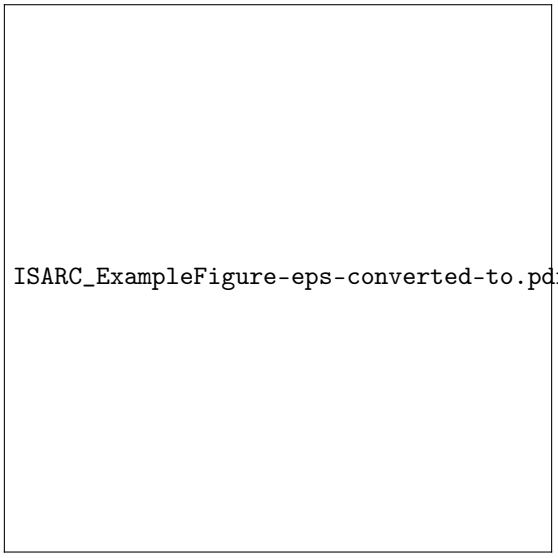
Figures should appear as close as possible to their first citation. They should be numbered consecutively using Arabic numerals and their title should be centered below the figure.

Figures, such as graphs and diagrams, should be embedded in vector format (e.g.`.eps`, `.pdf` or `.emf`), if at all possible. Otherwise, figures such as pictures should be with high resolution (300 dpi) when published at 100% (e.g., images at 72 dpi are in reality 25% of the required resolution).

For example, Figure 1 in this manuscript is embedded as a picture with high resolution (but, ideally, such graph should be prepared and embedded in vector format). Lines and lettering must be large enough (minimum 0.35pt thickness) to remain clearly legible when printed at 100%. For maps and similar figures, be sure to place a scale marker on the picture or photograph. Color is preferable; any grayscale figure requires sharp contrast. Please, refrain from using frames around figures, or using shaded backgrounds. Figures can significantly increase the size of the electronic file. Excessively large files tend to complicate and slow down the editing process. Therefore, you must make every effort to reduce the size of the electronic files of your figures. Embedding figures in vector format, as mentioned above, can significantly reduce file size while simultaneously enhancing figure quality.

The caption should be formatted with the word 'Figure' followed by the figure number, a period, a space, and the title. Each figure and caption should be separated from the adjacent text with one blank line. To achieve all this formatting, simply use the environment `figure` as in Figure 1.

A figure too wide to fit in one column may be placed across the width of the page (i.e. between the margins) by using the environment `figure*`. A figure too wide to fit between the margins may be placed in landscape orientation (sideways format), on a page by itself, with the bottom of the figure to the right of the page. To achieve this, embed the environment `figure` within an environment `landscape`.



Gambar 1: Effect of iron concentration on the amount of precipitate formed during hydrolytic precipitation from waste processing solutions

2.4 Tables

Tables should appear as close as possible to their first citation. They should be numbered consecutively using Arabic numerals and their title should be centered above the table.

The caption should be formatted with the word 'Table ' followed by a character space, the table number, a period, another character space, and the title. Separate each table from the adjacent text with one blank line. See Table 2 for example.

Table-wide lines (horizontal 0.5 point thickness) separate the title from the column headings, the column headings from the body of the table, and the table from the following text. Avoid vertical lines and avoid the use of horizontal lines between the various rows of data.

Text in the tables should have the same size as the main text, but not be indented and with space before and after it. Table 2 shows a good example. A table too wide to fit in one column may be placed across the width of the page (i.e. between the margins) by using the environment `table*`. A table too wide to fit between the margins may be placed in landscape orientation (sideways format), on a page by itself, with the bottom of the figure to the right of the page. To achieve this, embed the environment `table` within an environment `landscape`.

Tabel 2: Electron microprobe analyses of sphalerite grains in the Kidd Creek "C" concentrate

Element	Average(wt %)	Range(wt%)
Zn	60.8	59.6 - 63.3
Fe	5.82	3.54 - 6.95
Cd	0.30	0.12 - 0.42
S	3.31	33.6 - 33.5

2.5 Cross-referencing headings, numbered equations, figures and tables

Refer to a figure as 'Figure X', not using its relative position. For achieving this effectively, use the command `\label{}` within the environment `figure` to label each figure, and then use the command `\ref{}` in the text to refer to the label.

The same guidance applies to Tables and Equations, as well as Headings. In the case of headings, the command `\label{}` must be declared right after the heading commands (e.g. after the command `\section{}`).

Note that all the figures, tables, equations and the main headings in this template all have a label and are referred to in the text using the command `\ref{}`.

2.6 References

References to the literature should be cited in the main text with an Arabic number in square brackets [**?**]. References should then appear in cited order at the very end of your paper in the Section **References**. Start each reference on a new line with its number in square brackets [**?**]. Citation formats are given below for: a journal article [**?**], conference paper [**?**], book [**?**], and a website [**?**].

The formatting of the references is done automatically by the LaTeX template. All you need to do is add your references in the ISARC.bib Bibtex file and cite the papers in the text using the command `\cite{}` (as well as `\citet{}`).

Within the Bibtex file, please ensure that all references that have a DOI actually include it. It will then appear in the compiled references.